

Mapping the Legal Framework in Sri Lanka to Counter Cyber Terrorism- A Critical Analysis of Gaps and Enforcement Challenges

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by

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DECLARATION

We do hereby declare that the work reported in this dissertation was exclusively carried out by us under the supervision of Ms. C.S.D. Ellepola and Dr. Aparrajitha Ariyadasa. It describes the results of our own independent research except where due reference has been made in the text. No part of this dissertation has been submitted earlier or concurrently for the same or any other degree.

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Abstract

Cyber terrorism attacks constitute a growing risk to Sri Lanka's key services such as power, transport, water, banking, and telecommunications, since the country is heavily adopting digital technologies. In this context, Sri Lanka's readiness is evaluated by examining national laws and insights gained from cybersecurity professionals, academics, and Government Officials. In here maps four key legislations the Computer Crime Act (2007), Online Safety Act (2024), Data Protection Act (2022), and Electronic Transactions Act (2006) against eight global cyber terrorism trends. While they still have some legal gaps when covering these attacks.

Expert insights were gathered using stakeholder interviews and gathering insights from university academics, Cybersecurity professionals in the field, government officials, and legal practitioners. Academics urged clearer policies, greater international collaboration, and public awareness campaigns. Cybersecurity experts mentioned emerging threats, staffing shortages, and budget reductions, while law enforcement mentioned radical gaps in digital forensics capability and international cooperation.

In 2025, real experiences of cyberattacks on the Department of Pensions, Cargills Bank, and the National Water Supply and Drainage Board happen. Recently, a cyber attack happened to the Sri Lankan Ministry of Health database, through this attack, highly exposed patients sensitive personal identifiable information and leaked health records to the dark web. These attacks illustrate how weaknesses in legal and operational areas can break down services and inconvenience people.

We recommend legislation to encompass cyber attacks, upgrading cyber forensics training and facilities, keeping updated public training schemes continuously, and creating formalized channels of information sharing between the government, industry, and foreign partners. These measures will help Sri Lanka develop a more secure digital infrastructure to defend its critical services against cyber attacks.

Keywords: Critical Infrastructure; Cyber Resilience; Legislative Gaps; Stakeholder Insight; Cyber Terrorism; Public–Private Collaboration.

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List of Abbreviations

CNII	Critical National Information Infrastructure
DDoS	Distributed Denial-of-Service
GDPR	General Data Protection Regulation
SLCERT	Sri Lanka Computer Emergency Response Team
UK	United Kingdom
UNTOC	United Nations Convention Against Transnational Organized Crime

Introduction

This qualitative research project has used empirical methods to examine laws that counter cyber-terrorism broader framework of cyber warfare, focusing on Sri Lanka. The study examines the effectiveness of the cyber laws of Sri Lanka in protecting against cyber-terrorism. The research looks into Sri Lanka's current law on combating cyber-terrorism, where it determines what works and what needs to be corrected and enhanced. The study is particularly keen to observe Sri Lanka's cyber law against cyber-terrorism in analyzing both weaknesses and strengths in mapping the current legislation that is supposed to curb the same.

Furthermore, International frameworks Convention Against Transnational Organized Crime United Nations Convention Against Transnational Organized Crime (UNTOC) which enables countries to collaborate against organized crime that includes cyber-crimes, the International Convention for Suppression of Terrorism Financing making financing terrorism including cyber financing an offense, the SAARC Regional Convention which helps South Asian states to collaborate against terror, the International Convention for Suppression of Terrorist Bombings which includes terrorist bombing of public spaces and extends to cyber-attacks on critical infrastructure, and the Budapest Convention on Cybercrime which establishes guidelines for combating cybercrime need to follow as a country.

The research aims to find loopholes in the current cyber laws in Sri Lanka, suggest how Sri Lanka can improve its legal system by emulating international conventions as well as other countries, and help craft stronger, more versatile laws to tackle cyber-terrorism due to National security is the key point in the country.

1.1 Rationale for the Research

Cyber attacks, as well as cyber terrorism, present significant challenges to worldwide critical systems, particularly to developing countries like Sri Lanka that lack all-inclusive as well as coordinated legislation to combat these modern digital threats as well as cy-

ber vulnerabilities. One of the main reasons for these challenges is the quickly evolving nature of cyber threats, and knowledge of the hackers' high knowledge, which can advance much quicker than conventional systems of law can catch up. For instance, cyber terrorists incorporate advanced technologies like artificial intelligence, encrypted communications systems, as well as anonymous digital systems to conduct attacks that are designed to target critical systems like banking systems, power grids, as well as government networks. According unique characteristics, many existing laws created for traditional crimes are not suitable for dealing with cyber terrorism and cyber incidents.

Traditional legal frameworks, like Sri Lanka's Computer Crimes Act No. 24 of 2007 and Prevention of Terrorism Act No. 48 of 1979, were created prior to the digital age and are not always effectively equipped to protect citizens from cyber-terrorism. Though these laws deal with computer crimes to some extent, they neither specifically define the term "cyber-terrorism" nor offer stringent guidelines for implementation in the digital domain. These limitations may render weak legal measures ineffective in dealing with significant threats, particularly when cyber terrorists operate across international borders or use sophisticated technology to hide their activities. This is a significant challenge that needs to be addressed to enhance Sri Lanka's capacity to deal effectively with the sophisticated challenge posed by cyber terrorism in the modern world that is characterized by enhanced connectivity. This study seeks to enhance Sri Lanka's legislation effectiveness and obtain expert opinions from professionals by analyzing existing legislation, identifying gaps to deal with cyber-terrorism threats.

1.1.1 Context and current status of cyber-terrorism

Cyber-terrorism is turning into a significant national security, economic, and public security threat as the world gets progressively digital. Cyber warfare is defined as computer hacking for political motives in order to disable or destroy the capacity of an adversary. Sri Lanka, like any other country, is exposed to the risk that arises from attacks by cyber-terrorists on its critical infrastructure, communications, and public information systems. Whereas Sri Lanka has established various laws to combat cybercrime, Cyber-terrorism refers to attacks on digital systems for political reasons, aimed at disrupting or damaging targets like government websites, financial networks, and critical

infrastructure.

Cyber-terrorism is one type of cyber warfare in which hackers attack computer systems to harm infrastructure, spread terror, or achieve political gain. This includes attacks on digital systems for political motivations, aiming for government websites, finance networks, as well as critical infrastructure. These attacks may come from terrorist organizations or even nations themselves. Cyber-terrorists are often known to use tactics involving malware, phishing e-mails, as well as distributed Distributed Denial-of-Service (DDoS) in order to meet their objectives and fulfill their intentions. This review examines definitions, legislation, as well as challenges surrounding that of cyber-terrorism, particularly in Sri Lanka.

1.1.2 Sri Lanka's Legal Framework on cyber-terrorism

Sri Lanka works together with other countries under the Budapest Convention and UN treaties to fight cyber attacks. Sri Lanka has laws as the Online Safety Act No. 9 of 2024 and the Computer Crimes Act No. 24 of 2007 criminalize unauthorized access to computer systems, while the Prevention of Terrorism Act No. 48 of 1979 regulates traditional forms of terrorism. More recent laws, like the Data Protection Act No. 9 of 2022, control personal information. The Prevention of Money Laundering Act No. 5 of 2006 regulates criminal money use, though enforcement is insufficient. Professionals observe that these laws are not sufficient in relation to fighting cyber-terrorism, especially terrorist funding in this digital world. For example, mobile apps like "Binance" are still available in Sri Lanka without geo-blocking, while in other countries they are banned.

1.1.3 Challenges by the gaps of the legal literature and enforcement

Although Sri Lanka has made progress in regulating cybercrime, there seem to be huge loopholes in the legal framework for curbing cyber-terrorism. Enforcing laws against cyber-terrorism is difficult due to the hybrid nature of the terrorist attacks. Even though Sri Lanka has improved in governing cybercrime, gaps exist in large numbers in its laws against cyber threats. Enforcing laws against cyber-terrorism is difficult due to the hybrid nature of the terrorist attacks. Technology is always changing, rendering laws obsolete, with gaps to exploit. There are also no specific laws to regulate terrorist

activity on social media, gaming apps, and internet networks, thus leaving these spaces unregulated. There is no safety for public as well as private entities, especially critical infrastructure.

1.2 Problem identification

Rising cyber attacks and, due to its critical location in South Asia, require robust laws against cyber-terrorism. Currently, the Computer Crimes Act No. 24 of 2007 addresses general computer crimes but is silent on cyber-terrorism in particular, evidencing a manifest legal gap. Sri Lanka signed numerous international conventions, this, though, has not entirely implemented robust counter cyber-terrorism laws. Current initiatives, such as establishing the Cyber Security Agency of Sri Lanka, intend to augment the national cyber defenses, yet concerns remain regarding human rights versus security. Cross-country comparisons of Sri Lanka's legislation to model bills from other nations enable this country's approach to address cyber attacks to know where to improve. The research question is whether cyber-terrorism is an emerging threat that is threat to national security, whereas Sri Lanka may not have sufficient or coordinated legal frameworks to meet these new challenges. The study will critically analyze existing cyber laws to ascertain if these laws entirely deal with cyber-terrorism and identify gaps.

1.2.1 Research Questions

1. What are the key legal instruments currently in place in Sri Lanka to address cyber-terrorism?
2. How effective are Sri Lanka's existing legal frameworks in preventing and mitigating cyber-terrorism?
3. What gaps exist in Sri Lanka's legal frameworks with respect to emerging cyber-terrorism threats?
4. How do Sri Lanka's cyber security laws compare to international standards and best practices in addressing cyber-terrorism?
5. What specific legal and regulatory reform interventions are necessary to strengthen Sri Lanka's defense against cyber-terrorism? Apart from that, this study will be significant for policymakers, legal practitioners, and law enforcement agencies.

1.2.2 Relevance and Significance of the Study

Sri Lanka immediately needs strong cyber laws because today's global tensions, like this year's strikes with countries, show how groups use digital platforms for propaganda, recruitment, and attacks. As cyber threats overlap with national security, mapping and improving Sri Lanka's legal framework is vital to protect critical infrastructure and citizens from emerging vulnerabilities. This project will review existing laws, assess how well they work, and guide policymakers on needed reforms. By placing Sri Lanka's laws in the wider context of global security challenges, it shows why a flexible and unified legal approach is urgently needed.

There are various reasons to undertake this study. Firstly, as cyber attacks increase, understanding and strengthening Sri Lanka's legislation is integral to national safety and security. Identifying gaps will yield an improved policy to combat cyber risks. Secondly, in the global environment of growing tensions, as seen in the Israeli strikes, this research sheds light on the value of preparing for digital risks as much as physical ones. Next, the study will contribute to scholarly debates on cyber-terrorism as well as provide impetus to global cooperation through making Sri Lanka's legislation conform to standards as the Budapest Convention on Cybercrime. Overall, this study will enhance Sri Lanka's cyber resilience, safeguard its citizens, and enable the country to take a significant role in combating cyber-terrorism globally.

1.3 Research Objectives

1. To map and analyze the legal framework in Sri Lanka's countering cyber terrorism.
2. To evaluate the strengths and weaknesses of existing laws which curb cyber-terrorism in Sri Lanka.
3. To identify gaps in legislation and recommend areas for legal and regulatory reforms.
4. To assess the effectiveness of enforcement mechanisms in countering cyber-terrorism.
5. To give recommendations and suggestions for the necessary legal reforms in Sri Lanka to Curb Cyber-terrorism.

Literature Review

2.1 Theoretical Background

Theoretical foundations for legislation against cyber terrorism are based on several related frameworks responding to the changing nature of digital threats in the broader national security and international cooperation contexts. Here is an overview of the conceptual frameworks that underlie the forensic approaches to responding to cyber terrorism according to perspectives from the critical infrastructure protection frameworks, in relation to principles for international legal harmonization.

2.1.1 *Cyber Terrorism Conceptual Framework*

Cyber terrorism is defined as "unlawful attacks and threats of attack against computers, networks, and the information contained therein when perpetrated to intimidate or coerce a government or its citizens in support of political or social agendas" (Denning, 2023). Here, the definition creates cyber terrorism as a politically motivated activity that separates it from conventional cybercrime.

Cyber terrorism has also challenged traditional legal frameworks, as cyber attacks can potentially bypass state borders and reside in a space in which attribution is difficult. (Rid,2013) places cyber warfare, as well as cyber terrorism, in the category of politically motivated hacking intended to disrupt or damage adversary capability.

2.1.2 *Critical Infrastructure Protection Framework*

The key cybersecurity legislation challenge is protecting critical national infrastructure from terrorism. Critical infrastructure is that which is defined in the NIST Cybersecurity Framework as Critical infrastructure are the critical systems and assets, physical and cyber, whose disruption or destruction has the potential to significantly affect national security, the economy, public health, or safety (NIST, 2018).

This strategy emphasizes the mutual connectivity among existing systems of infrastruc-

ture as well as the cascading effects that are potentially generated under cyber attacks. The European strategy to secure critical infrastructure, as necessitated in the NIS Directive, is also directed towards sector-wide sector responses in energy, transport, finance, as well as healthcare industries (ECOWAS Cyberportal, 2024).

2.1.3 Previous Work

International Legal Frameworks and Conventions

The Budapest Convention for Cybercrime was undertaken by the Council of Europe in the year 2001. The Budapest Convention is the first computer-related crimes treaty in the world. The globe's first international cybercrime agreement, the Budapest Convention set standards for harmonizing domestic law, upgrading investigation methods, and increased international cooperation among states (Council of Europe, 2001). The Convention refers to offenses such as unauthorised access, interference with data, system interference, and misuse of devices, which national legislation adopts as the standard.

The more recent trends involve adopting the Second Additional Protocol to the Budapest Convention that fortifies provisions on cross-border cooperation and electronic evidence exchange. These trends show the change in how it is increasingly understood that cybersecurity threats need multinational efforts that can keep pace with the development of technology.

Sri Lankan Legal Framework Development

The Sri Lankan approach to counter legislation against cyber terrorism has come in ancillary trade and legislation acts rather than coming in one all-encompassing statute. Computer Crimes Act No. 24 of 2007, which criminalises unauthorised access, interference with information, as well as systems interference (Sri Lanka Legal Framework, 2024).

However, this Act does not specifically define the offence of cyber terrorism, thereby leaving loopholes for combating politically motivated cyber attacks. It has also added the Online Safety Act of 2024 and the Data Protection Act of 2022 to also address regulation of content and protection of privacy respectively.

Enforcement Challenges and Institutional Capacity

Literature constantly reveals important setbacks in enforcing cyber terrorism laws. A lack of technical knowledge in law enforcement agencies, low digital forensic capacity, and lack of resources impair efficient application of laws in place (Muganda, 2021). The Sri Lankan CID, for example, struggles to deal with sophisticated cases of cyber terrorism from shortfalls in specialized knowledge and tools. The mechanisms for international cooperation, though agreed upon through numerous treaties, are hamstrung by practical challenges such as jurisdictional intricacies, divergent systems of laws, as well as political imperatives that hinder or slow cross-cooperative working (Fernando, 2020).

Research Gaps

The study shows that cyber terror laws are unclear. In comparison to traditional terrorism, which is physical, cyber terrorism takes place on the internet. This makes it challenging to differentiate it from other crimes like hacking or espionage. Because of this, the government may either misuse anti-terror powers or fail to respond to genuine cyber terror threats appropriately. In Sri Lanka and the majority of countries, the difference between cyber terrorism and cybercrime has not been clarified yet.

Technology is ahead of the law. New tools like AI, cryptocurrency, blockchain, and smart devices present new threats that are not covered under existing laws. One such gap involves stopping terrorist financing through online platforms. International cooperation also falls short in confronting cyber terror. Countries are confronted with time lags in sharing information, slow legal processes, and poor coordination. The problem is compounded by overlapping treaties and the fact that not all countries are signatory to significant agreements. The patchwork system allows cyber terrorists to exploit loopholes, making global responses less efficient.

Materials and Methods

The research design outlines the systematic plan for conducting this study, detailing the methods and approaches that will be employed to achieve the research objectives. For this study, a qualitative research design is proposed, utilizing a combination of legal analysis, document review, and stakeholder interviews. This approach is tailored to address the complexities of mapping Sri Lanka's cyber laws in the context of cyber-terrorism. The following sections outline the components of the research design.

3.1 Theoretical Background

3.1.1 *Interpretivism*

Interpretivism is significant in this research by laying emphasis on the need to work towards constructing an understanding of the social context and subjective meanings of cyber laws in Sri Lanka. The theory believes that legal documents do not exist in a vacuum but rather are interpreted and understood in light of some cultural, historical, and social contexts (Schwandt, 1994). Using an interpretivist approach, this study will explore how various actors, such as legislators, law enforcers, civil society, and the general public, perceive and interpret existing cyber-terrorism legislation. This view allows for a critical examination of the implications of these laws on various communities, particularly on human rights and society's values, since legal systems can have mixed effects based on the social situation in which they operate (Mason, 2002).

3.1.2 *Pragmatism*

Pragmatism enhances this study with an emphasis on practical application and real-world applications of legislation. Pragmatism requires evaluating the effect of current legislation through observation of its solutions and outcomes as they translate to current issues (Dewey, 1938). Sri Lankan cybersecurity context, pragmatism requires a realistic examination of the implementation of existing laws and their ability to deal with the evolving nature of cyber threats. Based on different perspectives shared by various

stakeholders, the research aims to produce practical recommendations that can realistically be put into practice, thereby bridging the theory-practice gap. Such pragmatic action is in harmony with the need for legal frameworks to be adaptable and sensitive to the vulnerabilities of cyber-terrorism within a constantly changing technological landscape (Rorty, 1999)

3.1.3 Critical Realism

By recognizing the social structures and systems that exist below and within social practices and phenomena, Critical Realism serves as the underlying theory for this research (Bhaskar, 1998). This viewpoint makes it possible to examine the Sri Lankan approach to cyber-terrorism legislation, paying attention to the socio-political and economic factors that assist in its construction and implementation. The incorporation of critical realism in this research enables the analysis of powerful factors, institutional frameworks, and governance issues in the societal context that affect the legal frameworks for responding to cyber-terrorism and how these factors make the laws ineffective. This approach also enables the identification of the socio-political and institutional barriers for constructive action within the legal system, which makes it possible to propose criticisms and changes to improve the legal response to cyber-terrorism in Sri Lanka.

3.2 Research Approach

3.2.1 Qualitative Research

This research will adopt a qualitative research design to gain in-depth insights into the current legal frameworks for countering cyber-terrorism in Sri Lanka. Qualitative techniques are more appropriate for investigating legal concepts, understanding stakeholder perspectives, and establishing gaps within the existing laws (Creswell, 2013).

3.3 Research Methods for Mapping the Laws of Counter-Cyber-Terrorism Using R Studio

This research employs a comparative legal analysis methodology to map cyber laws for cyber terrorism. The study integrates qualitative research methods, leveraging R Studio for text mining, keyword searches, and data visualization to analyze data from legal texts and interviews. The research employs purposive sampling to select three expert

participants specializing in their area, ensuring the data collected is highly relevant to the research objectives.

3.3.1 Research Design

The study is designed as a comparative legal research approach, incorporating qualitative data collection and analysis methods.

- **Text Mining and Keyword Analysis using R Studio:** The qualitative data obtained from interviews, research documents, and legal texts will be analyzed using text mining and keyword analysis in R Studio, helping to identify recurring themes and concepts related to counter-cyber-terrorism laws.

3.3.2 Sampling Method and Data Collection Techniques

Purposive Sampling

- **Purposive Sampling** will be employed in choosing three experts in each category in their area of specialization. These experts will be selected from individuals with extensive knowledge and experience in the usage of cybersecurity laws, international treaties, and cyber-terrorism enforcement.
- **Justification for Purposive Sampling:** Purposive sampling is suitable since it aims at targeting the key stakeholders who are in the best position to offer in-depth information regarding the challenges of applying cyber laws and cyber-attacks. This will make sure the evidence obtained is of direct significance to the objectives of the study (Bryman, 2016).

Data Collection Techniques

In order to collect qualitative data, the study will utilize the following data collection methods:

- **Stakeholder Interviews:** Semi-structured interviews will be used in a way that will facilitate the in-depth exploration of the insights of the expert participants. Interviews will aim to gather more expert opinions on specific information regarding the enforcement policies of cyber laws, the problems of international

cooperation, and the loopholes or weaknesses in the current legal framework and cyberspace.

- **Flexibility:** Semi-structured interviews provide for the use of open-ended questions, which make it possible for interviewees to explain their answers in great detail, which might provide new information regarding the topic at hand, and critical areas of discussion are covered. Data Processing in R Studio: The interviews' transcripts will be processed in R Studio using packages such as tm and quanteda to conduct text mining, extracting frequent terms, concepts, and themes.

Interviews will be done with the following stakeholders.

No	Stakeholder Group	No of stakeholder
1	Legal practitioners	3
2	Scholars	3
3	Government officials law enforcement	3
4	Cybersecurity experts and Industry representatives	3
5	Military personnel	3

Table 1: Stakeholder Groups for Semi-Structured Interviews

Legal Texts and Documents

Legal documents, including national laws, will be collected for analysis.

Text Mining: The text mining of the legal documents will also be done using R Studio. It will involve extracting key phrases, terms, and legal clauses. Processing and data analysis will also be done efficiently using tm and quanteda R Studio libraries (Silge & Robinson, 2017).

Secondary Literature Review

A systematic literature review of recent literature will be undertaken for the purpose of establishing the academic articles, policy literature, and government reports on cyber-terrorism and counter-cyber-terrorism, and cyber-attacks will also be addressed.

Keyword Searches: Keyword searches will also be carried out with R Studio on the aggregated literature for the purpose of identifying dominant themes, gaps, and areas potentially rich for further work. Quanteda and tidytext will further be utilized for the tidying and exploration of this text data.

3.4 Data Analysis Methods

3.4.1 *Textual and Qualitative Analysis using R Studio*

- **Text Mining:** Text mining will be employed to process the data from legal texts and interview transcripts. The ‘tm’ and ‘quanteda’ packages in R Studio will be used for data cleaning, keyword extraction, and frequency analysis to identify recurring themes in the legal documents and interviews.
- **Thematic Analysis:** After text-mining, thematic qualitative data analysis will also take place, which will involve identifying and categorizing emerging themes from the interviews. Through this, there will also be an opportunity for the researcher to identify key areas of gaps in law, law enforcement challenges, and areas of improvement for policies.
- **Content Analysis:** Content analysis will be used to systematically evaluate the qualitative responses from the interviews. This will involve coding the data into meaningful themes or categories based on the content of responses (Saldaña, 2015).

3.4.2 *Visualization and Interpretation*

- **Data Visualization:** Visualizations such as bar charts, scatter plots, heat maps, donut charts, and pie charts will be created using R Studio to represent key findings from the interviews and text mining analysis. These visualizations will help highlight trends and patterns related to the enforcement and effectiveness of cyber laws.

3.5 Data Collection Techniques

- **Primary Data:** Collected through stakeholder interviews focusing on their experiences, opinions, and recommendations regarding cybersecurity laws by using a

purposive sampling method.

- **Secondary Data:** Collected from legal documents, academic literature, reports from governmental and non-governmental organizations, and relevant media articles.

3.6 Data Analysis

3.6.1 Content Analysis

- Content analysis will form the main approach to analyzing interview data, looking at how extant law responds to cyber-terrorism and where weaknesses. Following Braun and Clarke's (2006) approach, the analysis outlines recurrent themes and patterns, giving systematic insight into beliefs about current legal responses. Thematic analysis, due to its flexibility and rigueur (Nowell et al., 2017), is particularly appropriate to complex topics like cyber-terrorism. This type of analysis outlines strengths as well as weaknesses in the way that laws respond, as well as giving traction to later policy recommendations (Fereday & Muir-Cochrane, 2006).

3.6.2 Comparative and critical Analysis of Primary and secondary Data by using R

- This research undertakes a comparative and critical analysis of cyber laws using both interview data as well as legislative text, where R will provide the main software to process as well as analyze. Legislative text documents, i.e., statutes, will be brought into R by using the pdftools package in the case of PDF as well as plain text documents by the use of the tm or quanteda packages. Preprocessing will exclude irrelevant items (e.g., stopwords, punctuation), while keyword-in-context (KWIC) analysis will facilitate extracting significant terms such as enforcement, clarity, as well as compliance.
- For comparison, Latent Dirichlet Allocation (LDA) will be used to find thematic patterns throughout legal instruments, including in relation to clarity, applicability, enforcement, and coherence in responding to cyber-terrorism. Content and sentiment analysis will also be conducted on legal texts as well as interview con-

tent employing the syuzhet package to gain indicators on tone as well as framing.

- Furthermore, `textstat_frequency()` as well as `textstat_collocations()` will also be utilized to undertake frequency as well as co-occurrence analysis to identify how jurisdictions define as well as enforce the term cyber-terrorism. Statistical software, like regression analysis, will, where available, be utilized to undertake the analysis between enforcement practice in law, as well as use R Studio for this analysis.

Results and Discussion

4.1 Overview

This research investigates the Sri Lankan legal system to counter cyber terrorism by focusing on existing laws, enforcement mechanisms, and institutional capacities. It has been done on the basis of a comprehensive literature review, document analysis, and stakeholder interviews. It identifies the strengths, loopholes, and weaknesses in law and enforcement and proposes potential reforms to become more effective in countering cyber terrorism in Sri Lanka.

4.2 Analysis of Stakeholder Interviews

In this research, we gather expert insights from professionals in their expertise fields, including cybersecurity experts, legal practitioners, scholars, and government officials etc. We utilize purposive sampling to obtain these insights. By analyzing this information, we aim to identify the loopholes and challenges present in Sri Lanka's current cyberspace and legal framework. And also use thematic analysis to extract key points in their insights.

4.2.1 Thematic Analysis of Scholar Interviews

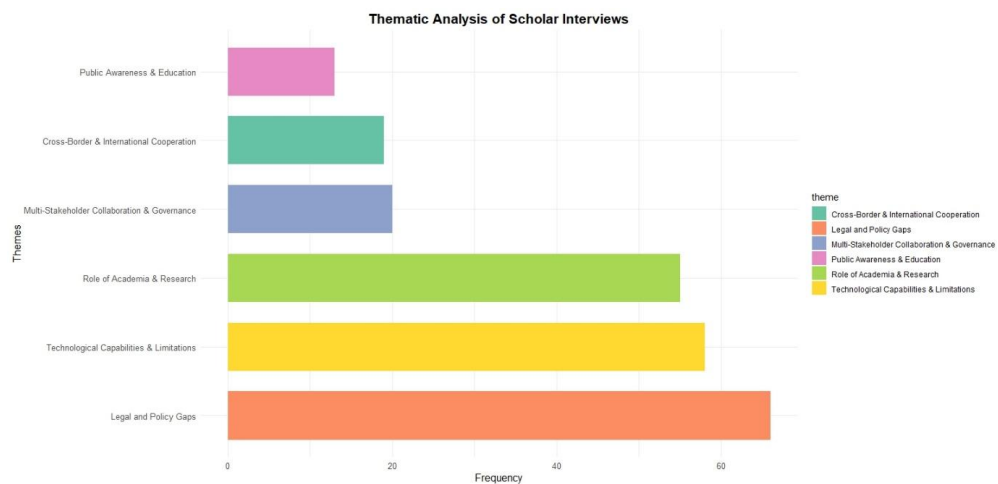


Figure 1: Thematic Analysis of Scholar Interviews

The thematic analysis of scholar interviews has highlighted essential critical themes for understanding the current landscape and future directions. These insights show the expertise of scholars engaged in policy discussion. By categorizing and assessing their frequency, we can pinpoint areas for immediate attention and collaboration.

In these interviews, participating professors and lecturers in the cybersecurity research area shared their insights, which were extracted as major contributions for this research. According to their expertise, the following key insights were identified.

Key Insights and Implications

According to academic professionals, immediate attention is needed for these issues:

- **Addressing Legal and Policy Gaps:** Governments and related organizations should update outdated laws and create flexible legal frameworks to handle emerging challenges in Sri Lanka.
- **Investing in Technological Advancements:** When new technologies are developed or emerge, there should be clear policies to improve cybersecurity, provide professionals with access to digital tools, and build capacity within organizations.
- **Enhancing Multi-Stakeholder Collaboration:** Scholars emphasize the need for stronger partnerships between governments, the private sector, and academia to create more effective solutions to enhance national security.
- **Strengthening International Cooperation:** Cross-national partnerships can facilitate harmonization of regulations, enhance global alignment on issues such as digital governance and trade, and improve mechanisms for effective information sharing.
- **Prioritizing Public Awareness and Education:** Public awareness campaigns and educational initiatives are necessary so that policy measures can be universally understood, enforced, and sustainably implemented.

4.2.2 Thematic Analysis of Cybersecurity Experts' Interviews

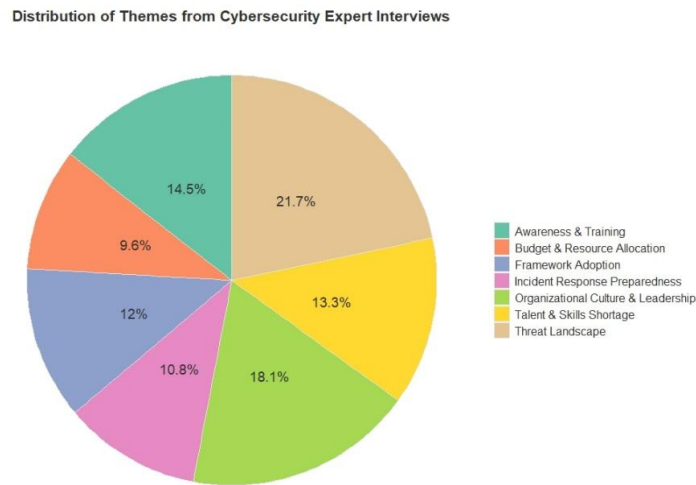


Figure 2: Distribution of Cybersecurity Themes from Expert Interviews

The thematic analysis of interviews with cybersecurity experts identified seven key themes that play a critical role in shaping cybersecurity strategies and policies. The diagram shows the distribution of themes and their relative significance based on cybersecurity expert insights. The findings suggest that although all factors are important, certain areas demand more immediate attention to enhance Sri Lanka's national security.

Key Insights and Implications

- **Threat Landscape:** Threat Landscape was the highly highlighted core theme of the analysis, and it was the subject of most of the expert interviews covering 21.7% of the responses. Experts emphasized that threats in cyberspace evolve by the day, become more sophisticated, and are even harder to detect and prevent. The attackers make use of the presence of vulnerabilities, ransomware, phishing, and state-sponsored cyber attacks. And need a strong solution for real-time monitoring, threat intelligence information sharing, and proactive security.
- **Talent & Skills Shortage:** Cybersecurity professionals and skill personnel shortage is a major problem, and 18.1% of responses, the thematic analysis shows. Cybersecurity experts noted that there are difficulties in hiring effective and skillful individuals by most organizations, which significantly difficult their capacity

to implement effective security measures, which affects their capacity to install effective security measures. As a country needs to invest in cybersecurity education, training programs, and workforce development in order to bridge this gap and prevent cyber breaches in the nation.

- **Awareness & Training:** Among the major issues brought up by specialists was the lack of cybersecurity training and awareness, which accounted for 14.5% of the discussion. Human error is responsible for most security breaches and cyber incidents, meaning that how much employee and public education on best cybersecurity practices is important. Organizations must highly prioritize cybersecurity awareness campaigns and training programs according to experts' insights to reduce the risks of cyber incidents.
- **Budget & Resource Allocation – Financial Challenges:** Budget limitations and resource constraints were mentioned by 13.3% of professionals as the main challenge to developing more robust cybersecurity defenses. The majority of organizations, particularly medium-sized enterprises (SMEs), struggle to invest sufficiently money in cybersecurity. Organizations and governments need to reallocate budgets to make cybersecurity an essential business investment rather than an optional expense.
- **Framework Adoption – Policy Gaps:** Cybersecurity policy and regulatory compliance took up 12% of the expert discussion. Some organizations are not fully aligned with cybersecurity best practices either due to a lack of regulatory enforcement or because the compliance requirements are complicated. Need simplification and enforcement of cybersecurity compliance policy by regulators for wider adoption.
- **Incident Response Preparedness:** In this section, detection, response, and recovery capabilities in the case of cyber incidents were mentioned in 10.8% of expert insights emphasized that a lack of a structured incident response in numerous organizations, and as a consequence, they remain vulnerable to prolonged cyberattacks. And also need to invest in threat detection systems, cybersecurity exercises, and incident response teams to minimize impact.
- **Organizational Culture & Leadership:** In this section mention leadership and

organizational culture are mentioned, accounting for 9.6% of feedback. Experts highlighted that cybersecurity had to be embedded in an organization’s culture, with leadership taking an active role in defining security policies. Organizations must develop a security-first mindset, where executives lead by example and promote a culture of cybersecurity awareness.

Overall Discussion & Insights

- **Balancing Priorities**

The study shows that while threat landscape and skills shortage top the charts, budget, training, and framework adoption are equally crucial.

- **Need for Proactive Strategies**

Experts highlight the need for a move from a reactive to a proactive cybersecurity approach with focus on preventive, rather than response, measures.

- **Collaboration is Critical**

Government, private sector, and academia must need to collaborate hands to develop stronger policies, conduct research, and incorporate cybersecurity training programs.

4.2.3 Comparative Analysis of Cybersecurity Practices by Industry

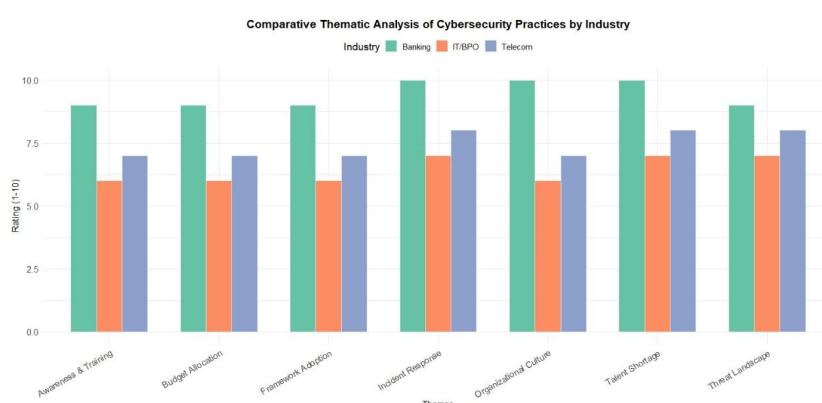


Figure 3: Comparative Analysis of Cybersecurity Practices by Industry

According to experts, cybersecurity practices differ from industry to industry depending on differences in budget planning, organizational management and goals, and regulations. The above chart graphically visualizes a comparative analysis of cybersecurity

practices in Banking, IT/BPO, and Telecom major sectors based on expert interview viewpoints. The industry shows different stages of cybersecurity preparedness, talent retention, and response capabilities. The comparison reveals strengths and weaknesses of different industries, showing industry-specific issues and areas of improvement. This table contains more data in relation to the diagram.

Theme	Banking (High)	IT/BPO (Low)	Telecom (Moderate)
Awareness & Training	Strong focus placed on training for cyber threat prevention.	Less spending on cybersecurity awareness.	Moderate training efforts.
Budget Allocation	Highest investment in cyber security infrastructure.	Limited budget, restricting security improvements.	Reasonable budget, yet inflexible financially.
Framework Adoption	High adherence to international security guidelines.	Having difficulties adopting best security practices.	Moderate implementation of the security frameworks.
Incident Response	Best responses for managing cyber attacks compared to other industries.	Lesser response capacity due to constrained finances.	Protocols are in place that are structured. Requires refinement.
Organizational Culture	Strong security culture with leadership involvement.	Cybersecurity is not a priority in company culture.	There are some focus on security but lack full integration.
Talent Shortage	Well-funded cybersecurity teams, minimal shortages.	Critical shortage of talent because of insufficient investment.	Moderate talent retention efforts.
Threat Landscape	Proactive stance using threat intelligence systems.	Less focus on monitoring evolving cyber threats.	Some awareness but lacks proactive strategies.

Table 2: Cybersecurity Readiness Assessment Across Different Sectors

After comparative analysis, key findings and recommendations have resulted in a more defined picture of industry-wide trends in cybersecurity. Strengths and challenges of

the industry and suggestions for greater industry-wide cybersecurity capacity.

Aspect	Findings
Industry Strengths	The banking industry has the highest spending on security investment, incident response, and employee awareness, and the telecom industry tries to align its security requirements with its budgetary constraints. This creates some challenges. In the meantime, the IT/BPO industry has some awareness of cybersecurity but lags behind because it has very limited spending on security investments.
Major Challenges	The IT/BPO sector has difficulty aligning budgeting against the need for cybersecurity, and the telecom sector requires stronger frameworks and more monetary investment for it to build more effective defenses. Across each sector, the common issue is in not having highly skilled cybersecurity personnel, and this has not yet been effective for protecting sufficiently.
Recommended Actions	Need more and more awareness of cybersecurity at every level. The IT/BPO industry requires improved planning for security, and the telecom industry needs more implementation of more effective infrastructures for security. While the banking industry is more advanced, it can also improve its policies for cybersecurity for more effective protection.

Table 3: Findings and recommendations

4.2.4 Cybersecurity Framework Adoption by Industry

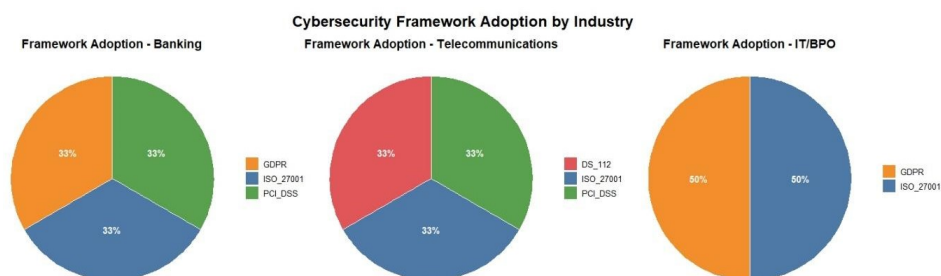


Figure 4: Cybersecurity Framework Adoption by Industry

The analysis of the adoption of the cybersecurity framework by three sectors, Banking, Telecommunications, and IT/BPO, shows various trends towards compliance options.

Utilization of each of these sectors differs on the scale of reliance on international guidelines and regulatory requirements.

Framework Adoption by Industry

- **Banking Sector**

- The banking sector has usage of General Data Protection Regulation (GDPR), ISO 27001, and PCI DSS frameworks equally, each being utilized to the extent of 33%.
- This reflects a top priority for data protection regulation (GDPR) and security management (ISO 27001), besides financial transaction security compliance (PCI DSS).
- The balanced approach shows the regulatory-driven security practice of the industry, coupled with the requirement for comprehensive protection due to the financial data being sensitive.

- **Telecommunications Sector**

- Communication companies also use the same 33% implementation rate of three different frameworks: DS 112, ISO 27001, and PCI DSS.
- ISO 27001 offers data security and confidentiality, while PCI DSS has shown a preference for payment-related data protection for telecom services.
- The presence of an industry-specific DS 112 is an example of sector-specific compliance based on telecommunications regulation.

- **IT/BPO Sector**

- Compared to telecom and banking sectors, IT/BPO companies identify their primary reliance in the form of GDPR (50%) and ISO 27001 (50%).
- The domination by GDPR suggests a strong focus on data privacy, particularly for companies handling foreign client data.
- The symmetrical split with ISO 27001 brings focus to the need for systematic security management in outsourcing and cloud operations.

Key Findings and Discussion

- **Regulatory Influence:** The banking industry compliance needs demand multi-framework adoption, while IT/BPO companies give utmost importance to GDPR and ISO 27001, possibly because of the international nature of their businesses.
- **Security vs. Compliance:** While every industry offers basic security with ISO 27001, banking and telecommunications include special regulations (PCI DSS & DS 112) that correspond to industry threats.
- **Industry-Specific Needs:** IT/BPO companies appear less transaction-oriented in their dependency on security (e.g., PCI DSS), as their industries are information-oriented rather than finance-oriented.

Findings indicate that the adoption of a cybersecurity framework is driven by industry requirements and regulations.

- Banking institutions balance multiple compliance frameworks to ensure financial and data protection.
- Telecommunication operators focus on network protection, confidentiality, and regulatory compliance.
- IT/BPO firms emphasize data protection and international standards, adapting to their outsourcing model.

4.2.5 Thematic Analysis of Legal Experts' Interviews

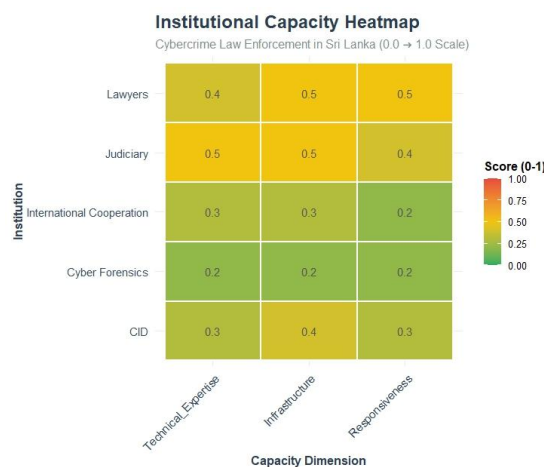


Figure 5: Institutional Capacity Assessment Based on Heatmap Analysis

The heatmap presents a comparative assessment of institutional capacities across different law enforcement and legal institutions in Sri Lanka, focusing on three key dimensions: Technical Expertise, Infrastructure, and Responsiveness. The scores range from 0 to 1, where higher values indicate stronger institutional capacity.

Institution	Technical Expertise	Infrastructure	Responsiveness	Key Observations
Lawyers	0.4	0.5	0.5	Moderate capacity, especially in infrastructure and responsiveness.
Judiciary	0.5	0.5	0.4	Stronger institutional knowledge but moderate responsiveness.
International Cooperation	0.3	0.3	0.2	Limited engagement in international collaboration.
Cyber Forensics	0.2	0.2	0.2	Weak capacity across all three dimensions, indicating the need for investment.
Criminal Investigation Department (CID)	0.3	0.4	0.3	Moderate infrastructure but lacks strong technical expertise.

Table 4: Findings and recommendations

Key Findings and Discussion

• Legal and Judiciary Capacity

- According to the heatmap, lawyers and the judiciary were ranked compare to high (0.4 - 0.5) in all three mentioned dimensions, showing moderately strong experience in handling cybercrime cases by legal professionals in Sri

Lanka.

- However, the judiciary's responsiveness (0.4) is lower, suggesting that legal proceedings and cases may experience delays in addressing cybercrime incidents efficiently.

- **Challenges in International Cooperation**

- The lowest scores are for international cooperation (0.2 - 0.3) in every dimension. There is a lack of a structured mechanism model for cooperation in investigating cybercrime in a cross-border direction.
- Strengthening international cooperation agreements and frameworks is necessary to improve law enforcement effectiveness in handling global cyber threats.

- **Weaknesses in Cyber Forensics Capacity**

- Cyber Forensics had the least (0.2) in all areas, reflecting a serious gap in expertise, infrastructural support, and responsiveness.
- That implies a pressing requirement for sophisticated forensic technologies, training programs, and funding and resource allocation in order to improve digital evidence gathering as well as analysis.

- **Role of the CID in Cybercrime Investigations**

- The CID scores moderately (0.3 - 0.4) but lacks the strong technical expertise needed for it to perform ability in conducting cyber investigations effectively.
- Providing high-end cybersecurity training and advanced next-gen technologies facilities to improve their capacity and accuracy of investigations

Discussion and Recommendations

Based on the analysis, it is recommended the following improvement measures are recommended for cybercrime law enforcement capacity in Sri Lanka:

- **Enhance Legal Training & Awareness:** Offer specialized education and special training to prosecutors, lawyers, and judges on the latest trends of cybercrime and

the latest developments in the methods of digital forensics.

- **Improve International Collaboration:** Strengthen the bilateral and multilateral agreements with an aim of allowing smooth cross-border cybercrime investigation and information exchange.
- **Invest in Cyber Forensics:** Provide budget allocations for cyber forensics hardware and software, and provide expert-level skill development to improve the investigatory capacity.

4.2.6 *An Analysis of Strengths and Weaknesses in Sri Lanka's Cybersecurity Legal Framework - Legal Experts*

♣ **Legal Framework Coverage: Gaps & Strengths in Sri Lanka**

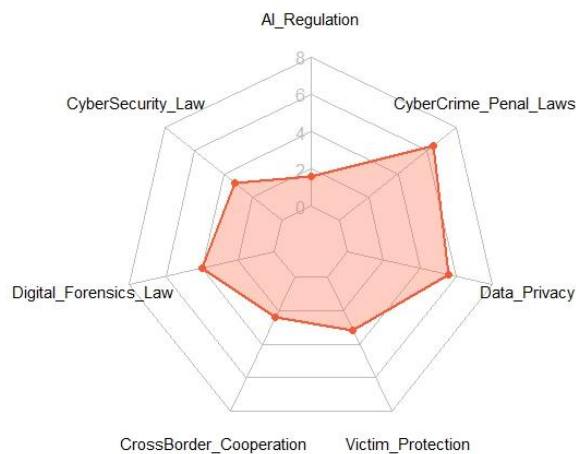


Figure 6: An Analysis of Strengths and Weaknesses in Sri Lanka's Cybersecurity Legal Framework - Legal Experts

The radar chart shows an illustration of weaknesses and strengths in Sri Lanka's cybersecurity legal framework in several key areas, including Cybercrime Penal Laws, Data Privacy, Digital Forensics Law, and Cross-Border Cooperation. These axes relate to different aspects of legal coverage in a manner such that higher values indicate better provisions in laws, while lower values indicate gaps.

Strengths in the Legal Framework

- Cybercrime Penal Laws and Data Privacy laws show comparatively better coverage compared to other areas.
- Cyber Law also evidences moderate coverage, suggesting some foundational policies are in place.

Gaps and Weaknesses

- Cross-border cooperation has weakened in coverage due to challenges in international cooperation in cybercrime investigations.
- Victim Protection is equally underdeveloped and points towards a requirement for robust policies for victims of cybercrime.
- Digital Forensics seems to be a weak link, implying minimal legal support for digital evidence gathering and forensic examination.

Discussion and Implications

- The failure to implement proper cross-border cooperation mechanisms may deter a response by law enforcement agencies to cybercrimes carried out in other countries. Upgrading cooperation and treaties can enhance response capacity.
- Insufficient protection for victims can lead to an insufficient support system for victims of cybercrime, requiring new policies for access to justice and victim support.
- Gaps in digital forensics indicate a need for legislative reforms to establish clear forensic investigation procedures for handling cyber evidence.

Recommendations for Improvement

- **Strengthen International Collaboration:** Develop a framework for international relationships to achieve effective cross-border information exchange and advanced cybercrime investigations.

- **Strengthen Victims' Protection:** Set up inclusive support mechanisms for cybercrime victims at the judiciary level with easily reachable legal assistance and specialized support service.
- **Improve Cyber Forensics:** Include in the statute special provisions for the systematic acquisition, preservation, and analysis of computer evidence to ensure admissibility and integrity at trial.
- **Expand Cyber Security Legislation:** Expand the coverage of existing cyber security policies to cover new threats, including cybercrimes and complex cyber-attacks.

4.2.7 Thematic Analysis of Government Officials' Interviews

Distribution of Stakeholder Responses - Government Officials

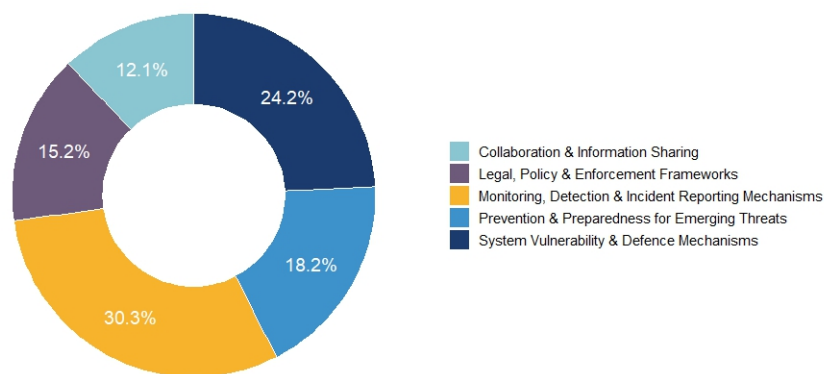


Figure 7: Distribution of Stakeholder Responses-Government Officials

The following chart represents stakeholder responses by government officials regarding Sri Lanka's cybersecurity action based on Sri Lanka Computer Emergency Response Team (SLCERT).

The chart highlights five major thematic clusters. Monitoring, Detection & Incident Reporting Mechanisms (30.3%) represents real-time monitoring, reporting mechanisms,

hotlines, and social media threat detection. System Vulnerability & Defence Mechanisms (24.2%) comprises server vulnerability testing, third-party audits, and threat detection technologies; Prevention & Preparedness for Emerging Threats (18.2%) covers national cybersecurity strategies, threats powered by artificial intelligence, and recurrent reviews of strategies Legal, Policy & Enforcement Frameworks (15.2%) reflects laws, policies, enforcement measures, and governance frameworks and Collaboration & Information Sharing (12.1%) capture public and private and international cooperation. Experts raised these issues that need immediate focus.

4.2.8 Distribution of Themes in SLCERT Officials' Responses

In moving to a more in-depth themes discussion, the chart provides a visual illustration of how stakeholders' interview responses were categorized into key themes.

Distribution of Themes in SLCERT Officials' Responses

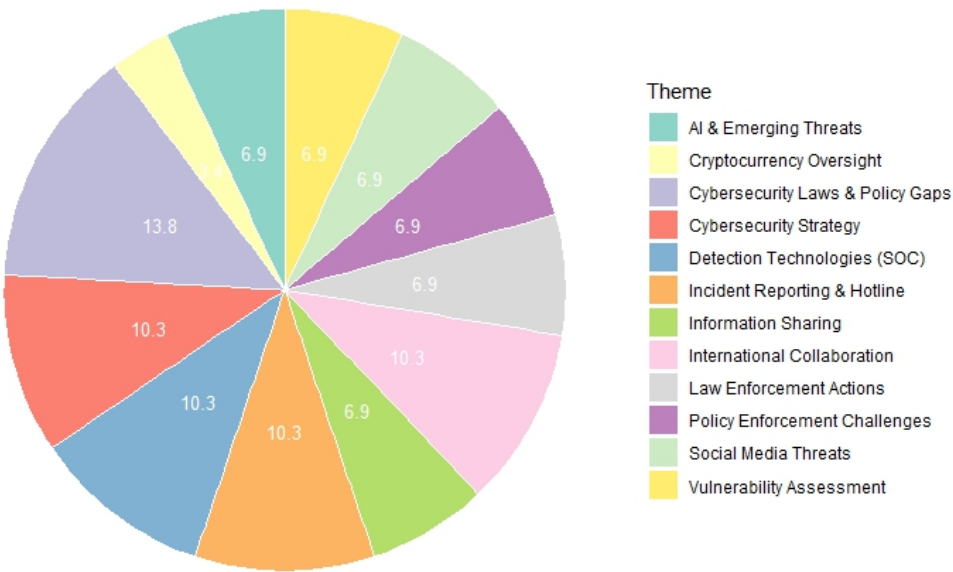


Figure 8: Distribution of Themes in SLCERT Officials' Responses

The most frequently discussed topic was Cybersecurity Laws & Policy Gaps, accounting for 13.3% of the total responses. This was followed by Cybersecurity Strategy, Incident Reporting & Hotline, and Detection Technologies (SOC), each comprising 10.0%. Themes such as International Collaboration and Information Sharing were equally represented at 10.0% and 6.7%, respectively. Emerging areas like AI & Emerging Threats,

Social Media Threats, Policy Enforcement Challenges, Law Enforcement Actions, and Vulnerability Assessment each contributed 6.7% to the total. The least discussed but still notable theme was Cryptocurrency Oversight, comprising 3.3% of the total. The chart visually emphasizes that legal frameworks and national strategy remain dominant concerns among government officials, while operational gaps in Cybersecurity readiness and enforcement mechanisms also require attention.

4.3 Analysis of Document Analysis

4.3.1 International Legal Frameworks and Cyber Security

Based on a comprehensive thematic analysis of the International Legal Framework and following is the breakdown in detail.

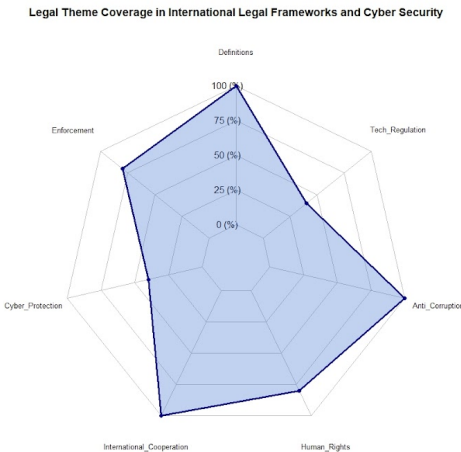


Figure 9: International Legal Frameworks and Cyber Security

The radar chart provides a visual representation of the degree to which the addresses seven key legal issues involved in the fight against cyber threats, including cyber warfare. The themes are rated 0 (no coverage) to 5 (strong coverage).

Moderately Covered Areas

- **Enforcement (4/5)**

While the international conventions and law encourage prosecution and enforcement, it does not specify digital investigation processes or methods for handling encrypted or online evidence that are central to modern cyberterrorism cases.

- **Human Rights (4/5)**

Outline protections for witnesses and victims. Little importance is given to digital rights, however, such as privacy while under surveillance or misuse of information on the internet

Overall Conclusion

These are major gaps that highlight the need for Sri Lanka to supplement existing laws with modern cybersecurity legislation. Explicitly define cyberterrorism within national legal instruments. This visual and thematic analysis reinforces the argument that Sri Lanka's legal framework must adapt to encompass the digital dimensions of terrorism. In light of the comparative evaluation between the United Nations Convention, international legal standards, and Sri Lanka's domestic framework, the findings underscore the necessity of timely and comprehensive legal reforms.

Recommendations for Sri Lanka's Legal Alignment

1. Adopt Cyber-Specific Procedures to fill the convention with modern cyber offenses.
2. Adopt Digital Witness Protection.
3. Expand Law Enforcement Mechanisms to include cyber forensics, access to encrypted evidence, and cross-border digital investigations.

4.3.2 National Framework and Key Legislation

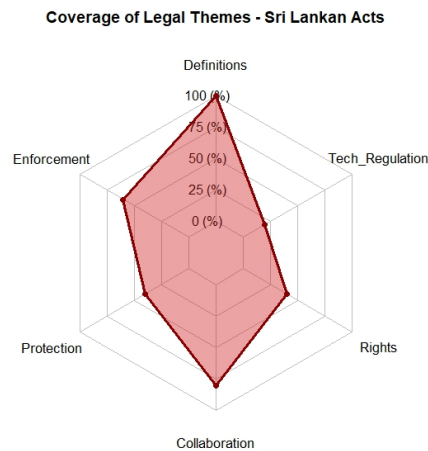


Figure 10: Coverage of Legal Themes - Legal Acts

Thematic Analysis of Sri Lankan Acts

- **Definitions & Scope of Cybercrime**

Sri Lanka's Computer Crimes Act No. 24 of 2007 criminalizes offenses such as unauthorized access, data interference, and device abuse. "Cyber terrorism" is not defined at all, and as politically motivated cyber attacks are faced, there are gaps in interpretation in applying the law.

- **Enforcement Mechanisms**

The Prevention of Terrorism Act grants broad powers of arrest, surveillance, and detention, but lacks cyber-specific investigative methods. Limited training in digital forensics and ineffective proactive detection of cyber threats worsen enforcement issues.

- **Cyber Protection & Infrastructure Security**

- Establishment of a Cyber Security Agency
- Protection of Critical National Information Infrastructure (CNII)
- Development of incident response, CERT, and digital forensics strength in relevant authorities.

- **International Cooperation**

Sri Lanka is a signatory to the Budapest Convention and participates in international cybercrime response forums. As well as the need for more international cooperation, which aligns with global cybercrime cooperation standards.

- **Human Rights & Privacy**

The Personal Data Protection Act No. 9 of 2022 enhances consumer rights and privacy. The Online Safety Act has, however, been criticized for undermining freedom of speech, arbitrary use, and a lack of transparency. And also the Insufficiency of balance between the protection of rights and the enforcement.

4.3.3 Academic and Policy Literature

Thematic Keyword Frequency from Academic Literature

The following bar chart shows the frequency of thematically significant keywords extracted from the academic literature on cyberterrorism.

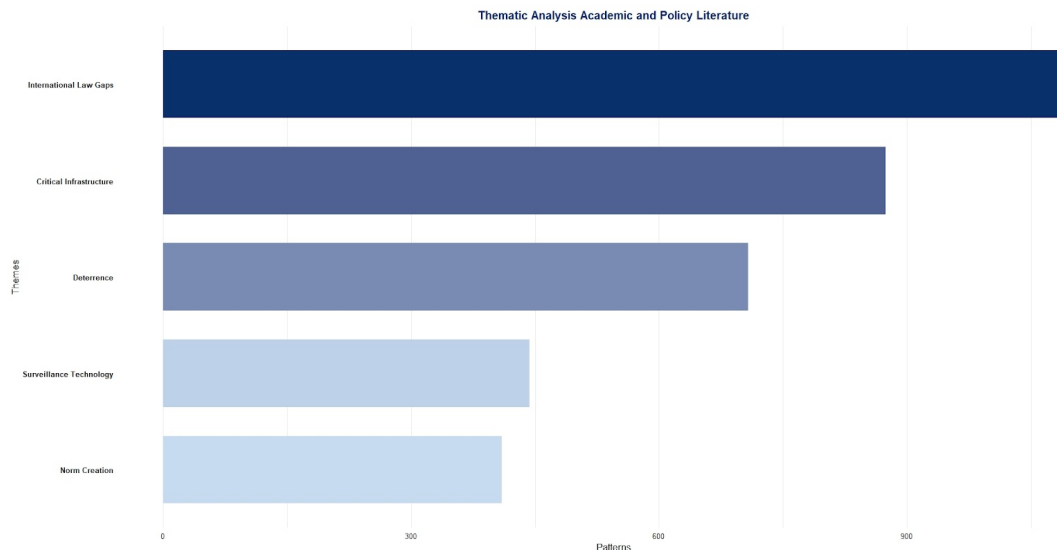


Figure 11: Thematic Keyword Frequency from Academic Literature

Deterrence was the most dominant theme, reflecting a shift of international strategy from legal norm-setting to the use of offensive cyber capacity.

Critical Infrastructure received a lot of attention, growing concern over vulnerabilities in sectors in the industry like power, water, and communication networks, to cyberterrorist

threats.

Norm Creation and Surveillance Technology were moderately represented. The literature shows that they do not possess enforceable norms or continue to be contentious due to privacy and civil liberties concerns.

This graph provides empirical support for deterrence and protection of infrastructure, leading to debate. The findings guide the argument of the research that the legal system of Sri Lanka must change in sync with international development, prioritizing cyber deterrence and infrastructure robustness, rights protection, and legal fragmentation.

4.4 Secondary Literature

4.4.1 Books and Monographs

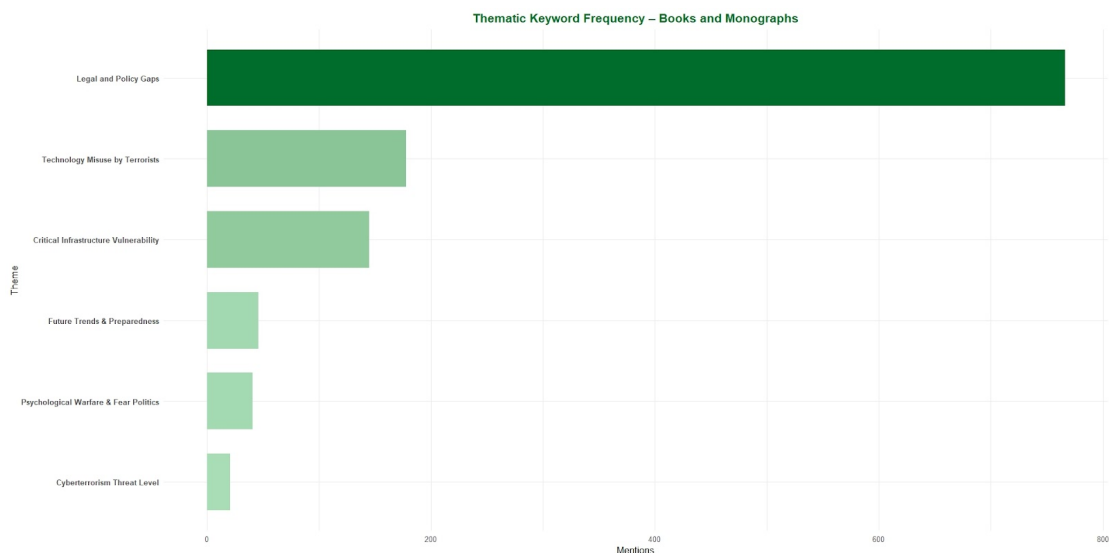


Figure 12: Books and Monographs

The bar chart shows the occurrences of keyword usage for every theme, providing a clear indication of the topics that have been most highlighted in the paper:

- Critical Infrastructure Vulnerability and Future Trends & Preparedness was given maximum focus, highlighting concerns regarding digital dependencies and future threats from more technically advanced terrorist actors.
- Fear Politics & Psychological Warfare was also very prominent, corresponding to the report's criticism of political and media coverage for overstating popular fear in the face of very rare instances of actual cyberterrorism.

- Legal and Policy Gaps demonstrated fair coverage, reflecting an awareness of weakness in regulations, but not a main focus. The visualizations reinforce the document’s policy-oriented nature, highlighting:
 - A strong focus on Infrastructure Resilience and Rising Risks
 - Continuing psychosocial impact of cyberterrorism narrative
 - A moderate between governance and legal development

These results can guide national policy discussions by showing from academic and institutional attention is concentrated and where further legal development is needed.

4.4.2 Journal Articles

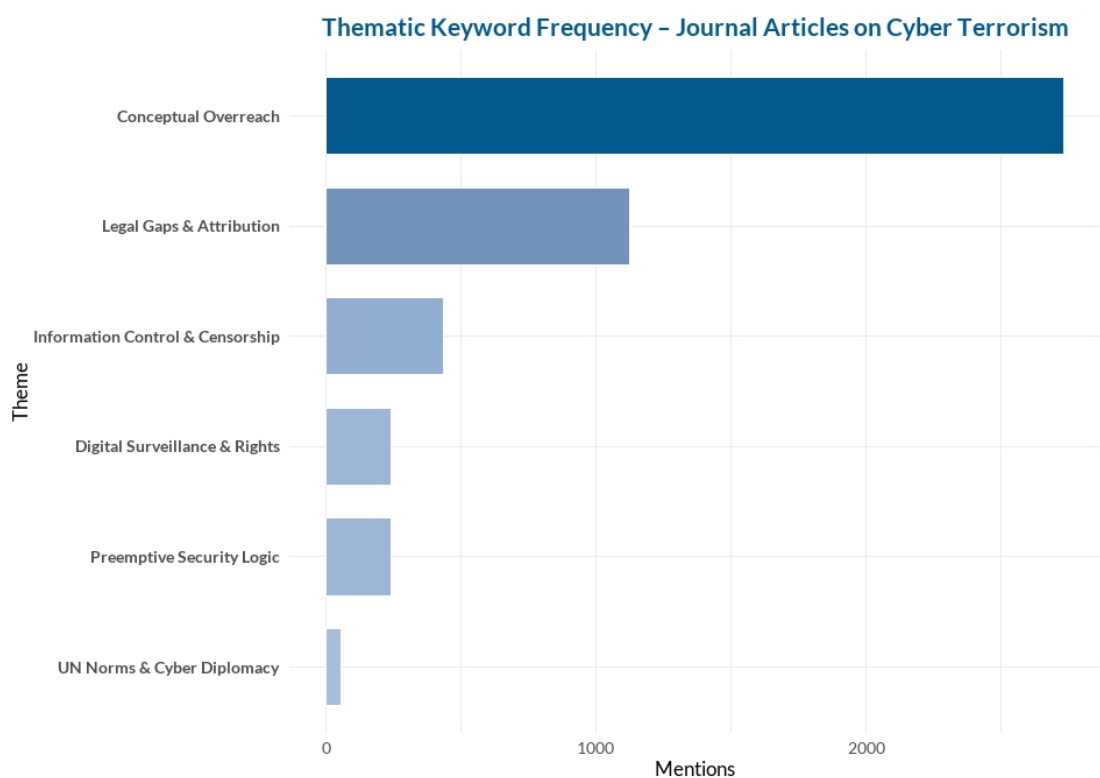


Figure 13: Journal Articles

Conceptual Overreach: Most frequently found themes, illustrating that the article strongly criticizes the broad and often vague term cyber terrorism.

Digital Surveillance & Rights: Critical examination of the way counter-terror policies

consistently lead to erosion of digital freedom, into mass-surveillance, surveillance of data, and a narrowing of civil space.

Legal Gaps & Attribution: Emphasis on the failure of international legal systems to properly identify, prosecute, or assign responsibility.

4.4.3 Reports and White Papers

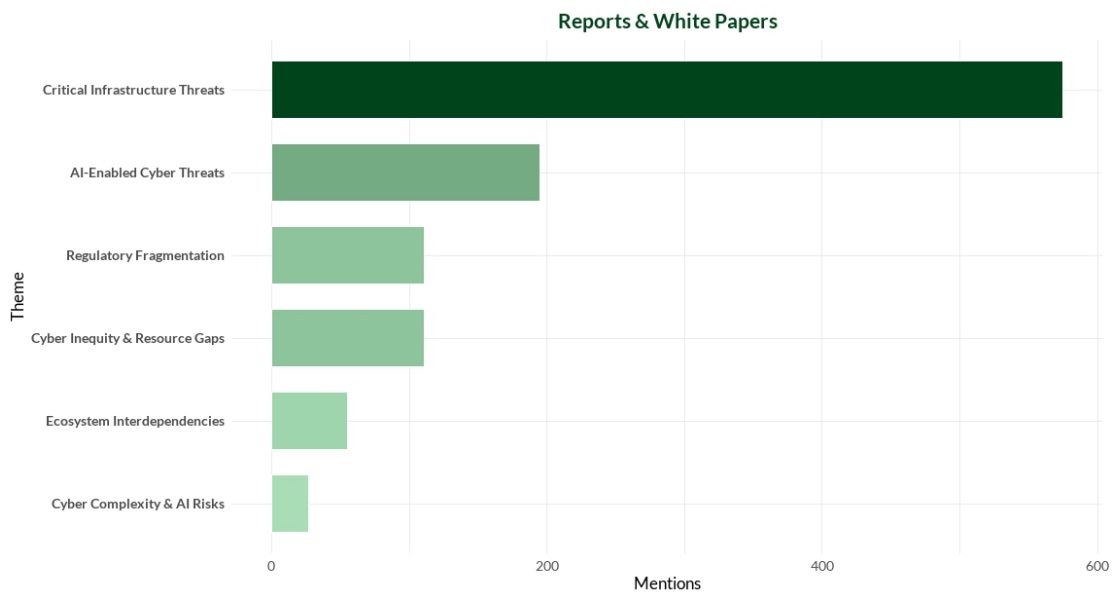


Figure 14: Reports and White Papers

- Cyber Complexity & AI Risks and Critical Infrastructure Threats appear as notable themes revealing the world's concern regarding cyber threats and attacks related to AI, as well as essential public services like communication systems, energy, and water.
- AI-Enabled Cyber Threats and Regulatory Fragmentation - a lot of concern about new threats from the generative technologies of AI and challenges from multiple jurisdictions.

4.4.4 Conference Proceedings

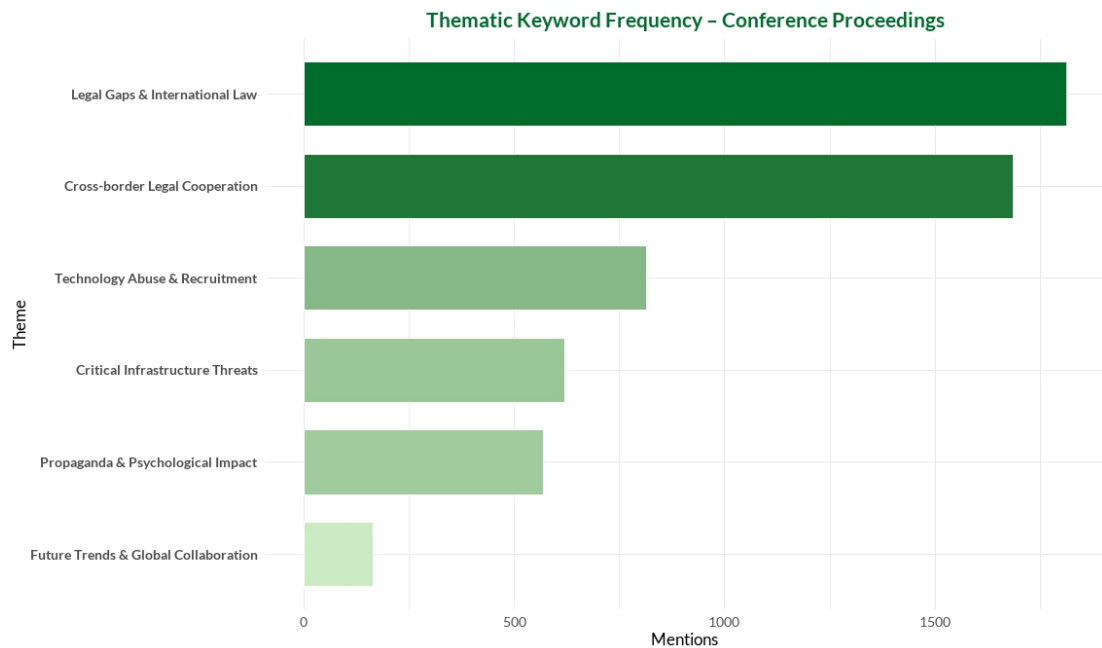


Figure 15: Conference Proceedings

- **Legal Gaps & International Law** emerges and revealing that the conference was concerned that laws today are insufficient for controlling and punishing cyberterrorism. This demonstrates that there is a persistent global concern that there are constraints upon legal power and differential reactions to cyber threats.
- **Critical Infrastructure Threats** is a significant topic, highlighting scholarly and policy recognition of how cyberterrorism targets essential public infrastructure such as energy networks, telecommunications networks, and emergency systems.
- **Propaganda & Psychological** impact are also thoroughly examined, emphasizing the role of cyberspace in spreading extremist political ideas, recruitment, and psychological fear. This ties together concerns about the social consequences of cyber attacks.
- **Cross-border Legal Cooperation** and **Technology Abuse & Recruitment** appear as interconnected themes, indicating ongoing discussion concerning the misuse of internet technologies in recruitment and the need for enhanced international cooperation.

4.4.5 Government and NGO Reports

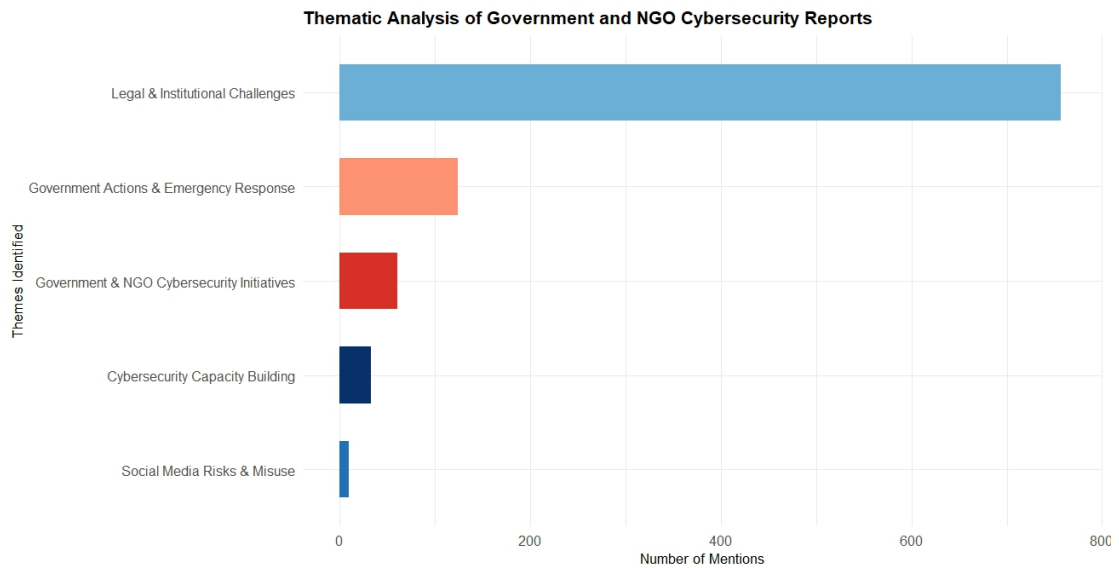


Figure 16: Government and NGO Reports

According to the analysis these points need to get immediate attention

- **Cybersecurity Capacity Building:** Includes awareness, education, and training programs to create national cyber readiness.
- **Social Media Risks & Misuse:** Focuses on disinformation, propaganda, and social media threats issues.
- **Legal & Institutional Challenges:** Gaps in laws and regulations, and enforcement mechanisms need to more focus this.
- **Government Actions & Emergency Response** Includes short-term responses like CERT responses, shutdowns, and national cybersecurity policies.
- **Government & NGO Cybersecurity Initiatives:** Focuses reports, proposals, collaborations, and governance structures from governmental and non-governmental organizations.

4.4.6 Dissertations and Theses

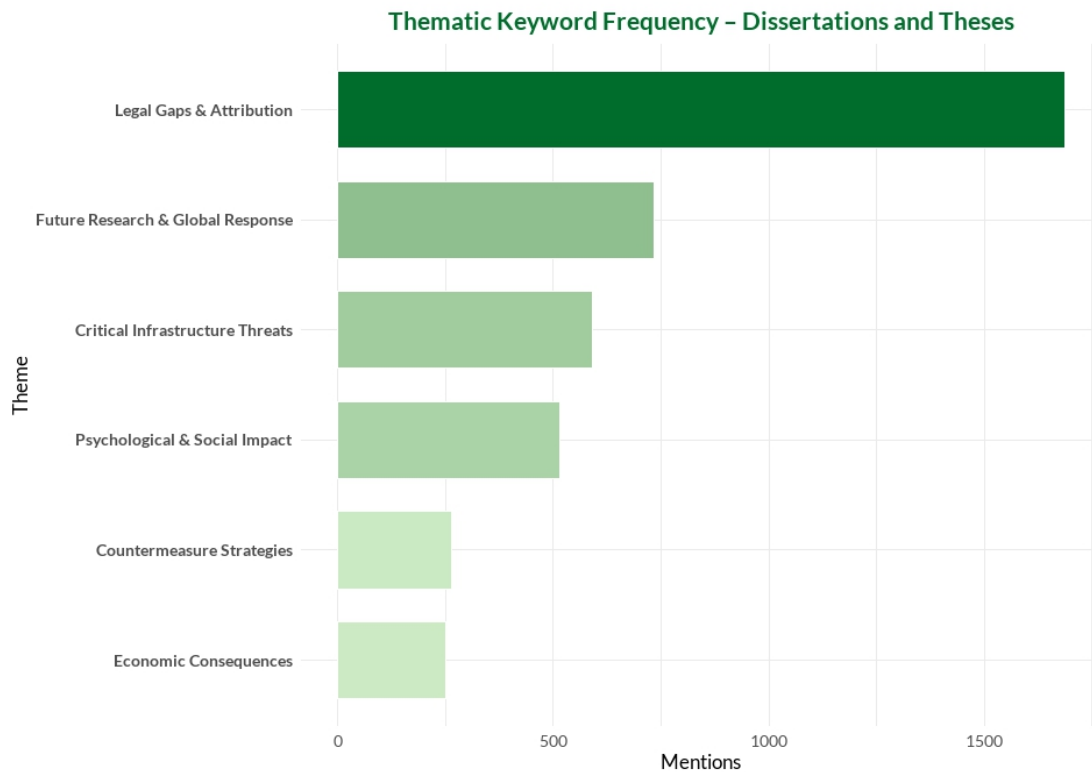


Figure 17: Dissertations and Theses

- **Legal Gaps & Attribution:** This theme was the most prominent, indicating a strong academic focus on the inadequacy of existing legal frameworks to address cyberterrorism.
- **Critical Infrastructure Threats:** In here, key public infrastructure such as energy grids, hospitals, water supply, and communications are under cyberattack and outlining the threat cyberterrorism poses to national security needs more attention.
- **Countermeasure Strategies:** This theme is characteristic of the fact that the thesis is not only diagnostic but also solution-oriented. The repetition of "firewalls," "preparedness," and "response".
 - Technical and institutional defenses
 - Public–private partnerships in cyber crisis management
 - National CERTs (Computer Emergency Response Teams) role

These are need to more enhance and powerful to face cyber incidents effectively.

4.4.7 Media Articles and Commentaries

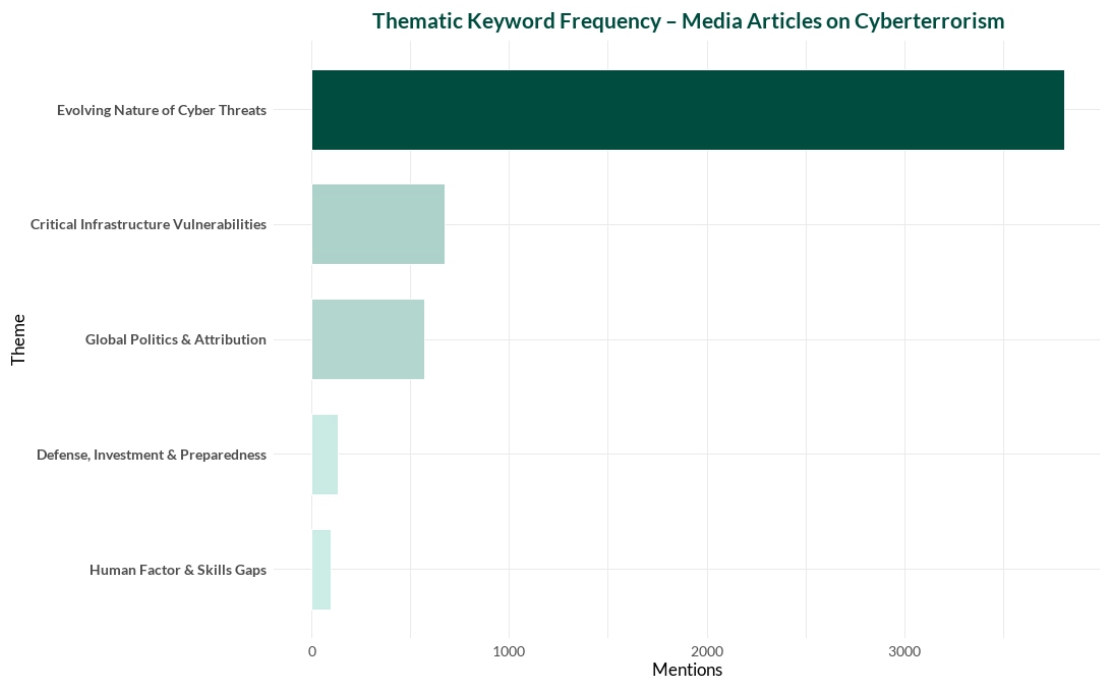


Figure 18: Media Articles and Commentaries

- **Evolving Nature of Cyber Threats:** Growing interest from the media towards the sophistication and rise of cyberattacks, including state-sponsored operations and espionage.
- **Critical Infrastructure Vulnerabilities:** Need to point out concerns about financial networks, power grids, and other vital public services being targeted by cyberattacks.
- **Defense, Investment & Preparedness:** Strong narrative of increased cybersecurity investments, especially in the banking sector, and proactive measures.

4.5 Mapping Legal Framework

Global Cyber Terrorism Trends The section below highlights some of the most notable global cyber terrorism trends, bringing together new threat vectors, tactics, and tools utilized by malicious actors worldwide. The highlights of the report, in a nutshell, are as follows,

Trend Area	Description
Youth Recruitment via Social Media	TikTok, Telegram, Discord, and in-game chat platforms are abused for the recruitment of youth. Propaganda is concentrated and distributed among the online avenues for targeting and radicalization of vulnerable individuals for terror activities.
Dark Web, Crypto Financing & Trafficking	Terrorist groups utilize Darknet markets and anonymous currencies (e.g., Monero, Zcash, privacy-focused DeFi protocols) for the laundering of funds, funding criminal activity, and online commerce. Blockchains conceal sources of funds, thus obscuring tracing.
Satellite & LEO (Low Earth Orbit) Infrastructure Threats	They encompass GNSS spoofing (altering satellite navigation signals), and satellite jamming. Satellite and LEO communications are also targeted more and more for disruption and eavesdropping.
Deepfake & AI-Driven Cognitive Attacks	Use of AI tools and deepfakes for psychological manipulation, spreading disinformation, triggering mental health crises (e.g., suicide gamification), and infrastructure sabotage. "Trauma engineering" employs AI to amplify psychological impacts and drive radicalization.
Hybrid Warfare, Digital Twins & Simulated Attacks	Cyber-physical simulated attacks on critical infrastructure such as power grids, ports, and hospitals have increased. Digital twins (virtual models of real-world systems) are used to plan and rehearse attacks before real-world execution.
Lone-Wolf Radicalisation & Social Media Weaponisation	Social media provides power for lone-wolf actors by virtue of ideological access and networks in a manner that evades traditional hierarchies. Telegram, TikTok, and YouTube provide for the spreading of extremist propaganda and self-radicalization, frequently by means of online gaming.
Gamification of Terror & Extremist Narratives	Discord and other online platforms, such as Roblox, are used for challenge-based indoctrination, including gamified methods of recruitment and member retention. These campaigns promote terrorism as a type of "achievement" or "challenge," particularly among youth.
Autonomous Weapons & Unmanned Vehicles	Increased employment of drone swarms, malware-based on artificial intelligence, unmanned ground and sea vehicles (UGVs, UUVs), UAVs, and automatic systems for sensing, delivery of cyber payload, and movement of explosives is seen.

Table 5: Global Cyber Terrorism Trends

4.6 Legal Coverage

To examine Sri Lanka's cyber laws in context to global cyber terrorism trends, the study first identified eight key areas of concern for global cyber terrorism risk from domain experts' findings. Key Sri Lankan cyber laws were then correlated in an organized fashion against these risk areas, examining each provision of the law's response or relation to the found threats.

The mapping procedure involved confirmation by professional consultation with lawyers to determine accuracy and relevance of the coverage examination. Comparison of each law to the global trends was quantified into ratio values, demonstrating relative strength or weakness of legal coverage in each element.

Finally, the established values were analyzed and represented by R Studio and the R programming language in terms of percentage-based graphs that illustrate the legislative focus distributed across the different cyber terrorism trends.

4.6.1 Sri Lanka Computer Crime Act- 2007

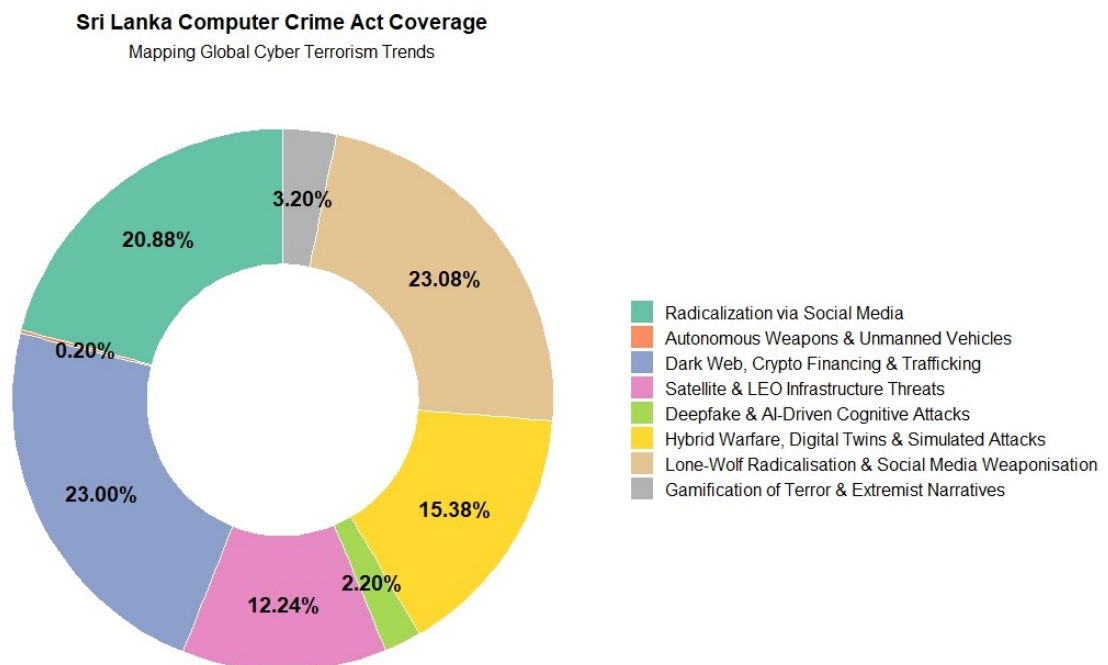


Figure 19: Sri Lanka Computer Crime Act Coverage

Sri Lanka Computer Crime Act Coverage Mapping Global Cyber Terrorism Trends provides a visual breakdown of how effectively Sri Lanka's Computer Crime Act addresses

each of the eight major areas of modern global cyber terrorism risk The Sri Lanka Computer Crime Act is more evenly but less balanced, directed at Global Cyber Terrorism Trends, with coverage distributed across all eight areas identified. The strongest emphasis is on lone-wolf radicalisation and weaponisation of social media (23.08%) and dark web, crypto funding, and trafficking (23%), which is a reflection of the Act’s strong focus against digital abuse and financial crimes. Moderate attention is given to radicalization via social media (20.88%), hybrid warfare and digital twins (15.38%), and threats to LEO and satellite infrastructure (12.24%), while low attention is accorded to gamification of violent extremist narratives (3.2%) and cognitive attacks based on AI (2.2%). Autonomous weapons and unmanned vehicles (0.2%) are accorded minimal attention, which represents an enormous gap in addressing next-gen technological threats. Overall, the Act provides broad coverage in most areas but is not as well-placed for novel AI and autonomous systems risks.

4.6.2 Online Safety Act

Sri Lanka Online Safety Act Coverage
Mapping Global Cyber Terrorism Trends

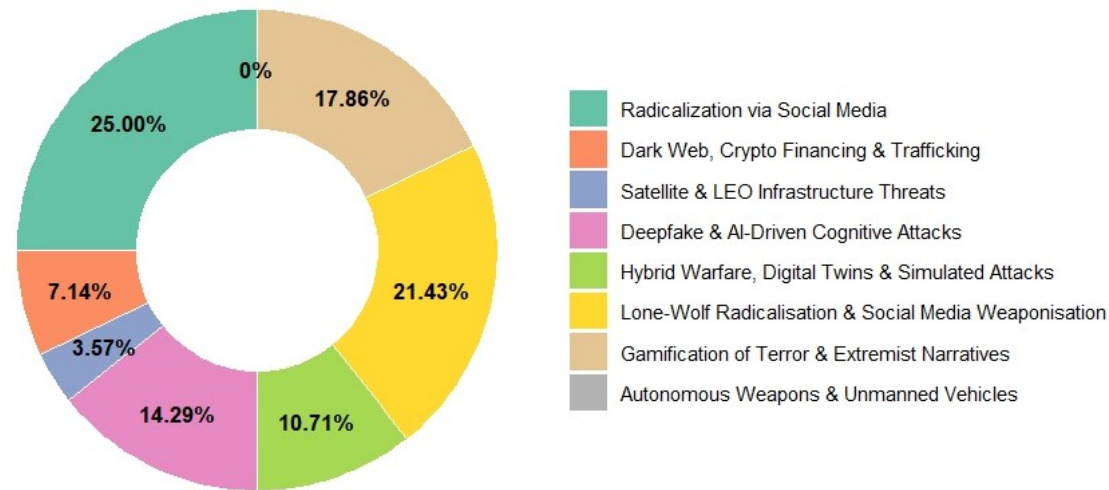


Figure 20: Sri Lanka Online Safety Act Coverage

Sri Lanka Online Safety Act Coverage (2024) Mapping Against 8 Global Cyber Terrorism Trends presents a quantitative assessment of how comprehensively Sri Lanka’s latest online safety legislation addresses contemporary global cyber terrorism threats.

The visualization displays normalized percentage coverage for each trend area, with segments proportionally sized to reflect the relative legal coverage provided by the Act. Sri Lanka Online Safety Act (2024) is an overall but skewed compliance with global cyber terrorism trends, offering legislative protection in 7 out of 8 of the core areas. The major focus is placed on Radicalization via Social Media (25%) and lone-wolf radicalisation (21.43%), highlighting the strong focus of the Act on social media-type threats. Moderate attention is given to extremist narrative gamification (17.86%) and cognitive attacks with AI (14.29%), while new topics such as hybrid warfare (10.71%), dark web and crypto-financing (7.14%), and satellite infrastructure threats (3.57%) are relatively less discussed. Importantly, autonomous weapons and unmanned vehicles are not addressed, which is a key legislative gap.

4.6.3 Data Protection Act No. 9 of 2022 Act

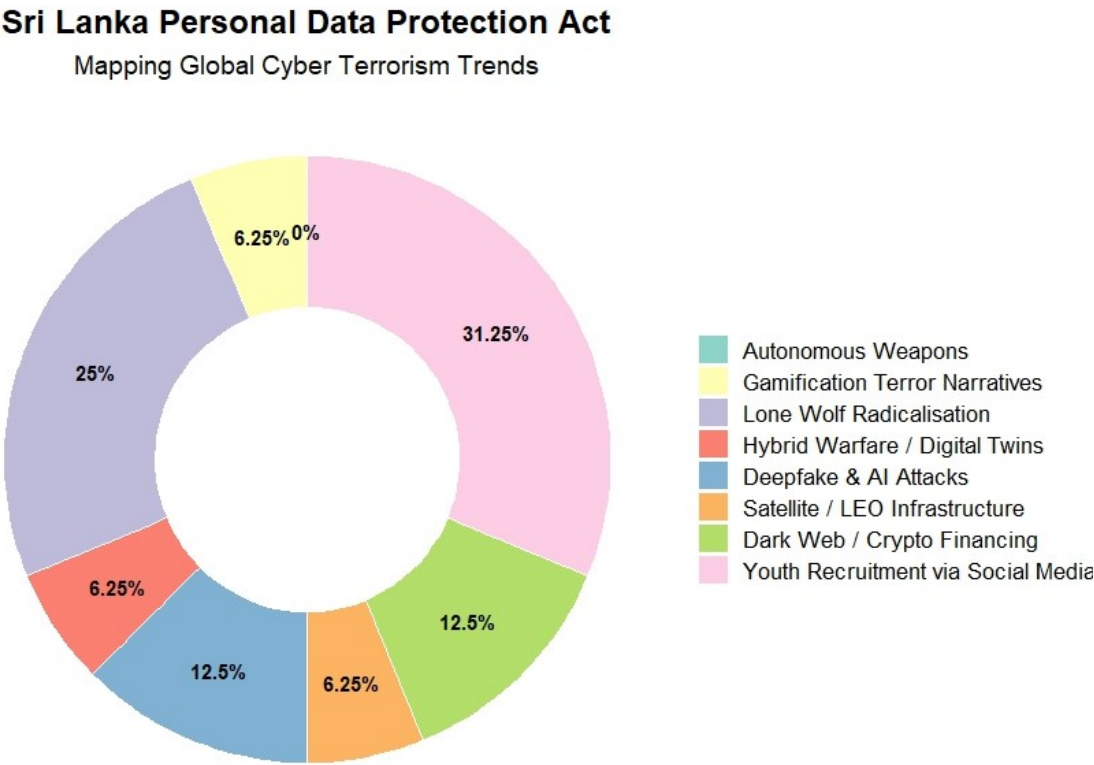


Figure 21: Sri Lanka Personal Data Protection Act

Sri Lanka Personal Data Protection Act (2022) addresses the eight identified Global Cyber Terrorism Trends. The area of each sector in the chart is directly proportional to the relative legislative coverage of each trend. The Sri Lanka Personal Data Pro-

tection Act of 2022 manages to cover part of the global cyber terrorism trends to a great extent through the protection of personal data and the regulation of data processing by law. The more robust area of intersections is social media-based radicalisation (31.25%) and lone-wolf radicalisation (25%), with prohibitions on unauthorised profiling and exploitative usage having indirect impacts. There is minimal coverage for hybrid war (6.25%), dark web and satellite infrastructure threats (6.25%), since there is a preference towards general data security provisions compared to special measures.

4.6.4 *Electronic Transactions Act, No. 19 OF 2006*

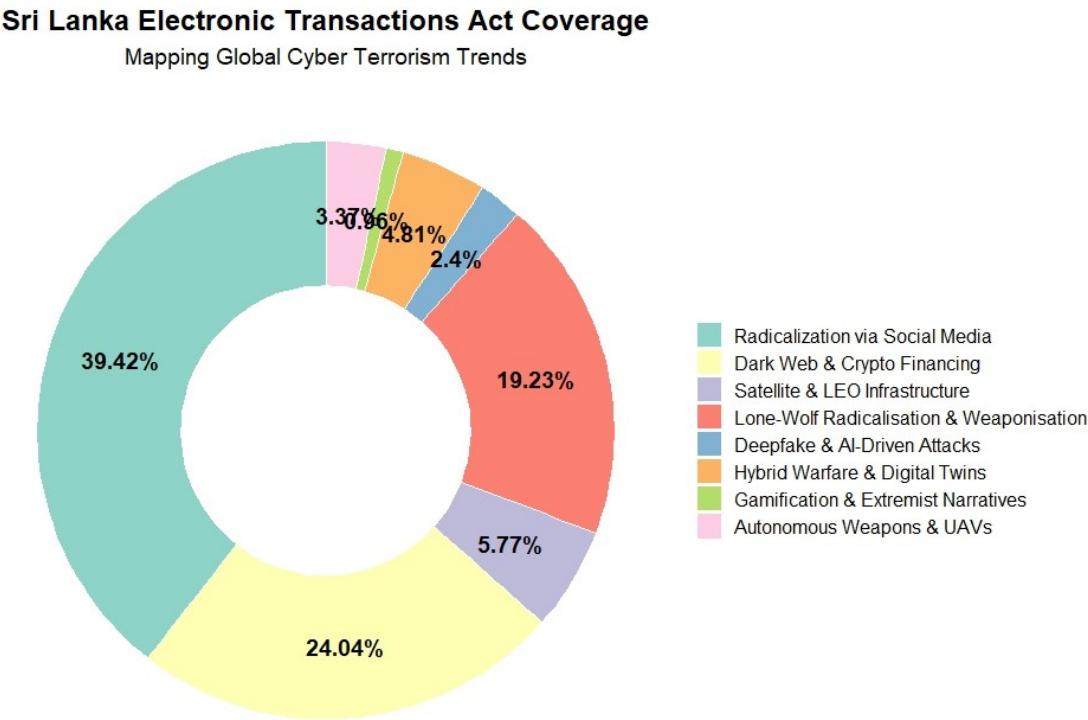


Figure 22: Sri Lanka Electronic Transaction Act Coverage

Sri Lanka Electronic Transactions Act (2006) represents half the exposure of emerging global cyber terrorism trends, primarily through legal acceptance of electronic communication, digital signatures, and encouragement of secure online payment. The Act mirrors its optimal concordance with social media radicalization (39.42%) by emphasizing authenticating messages and preventing fraudulent online activities, indirectly addressing extremist content distribution threats. Substantial coverage is also seen in crypto-financing and dark web (24.04%), given that the Act’s scope of regulation of electronic records, transactions, and enforceability by law of secure digital payments

can be used to thwart illegal online money transactions. Medium coverage is seen in satellite & LEO infrastructure threats (5.77%) and lone-wolf radicalisation and weaponisation (19.23%), where indirect protection is ensured through safeguarding against electronic fraud and unauthorised access. Less focus is on deepfake & AI-fueled attacks (3.85%), hybrid war and digital doubles (3.46%), and gamification of radical narratives (2.40%).

4.7 Comprehensive chart mapping the UK cybersecurity strategy

International Legal Frameworks and Cyber Security in the United Kingdom (UK). Based on the following documents, this mapping has been done.

- A Critical Analysis of the Gaps and Enforcement Challenges – National Policing Cyber Security Strategy UK.
- City Preparedness for Cyber-Enabled Terrorism: Literature Review Framework
- Government Cyber Security Strategy: Building a Cyber-Resilient Public Sector (2022–2030).

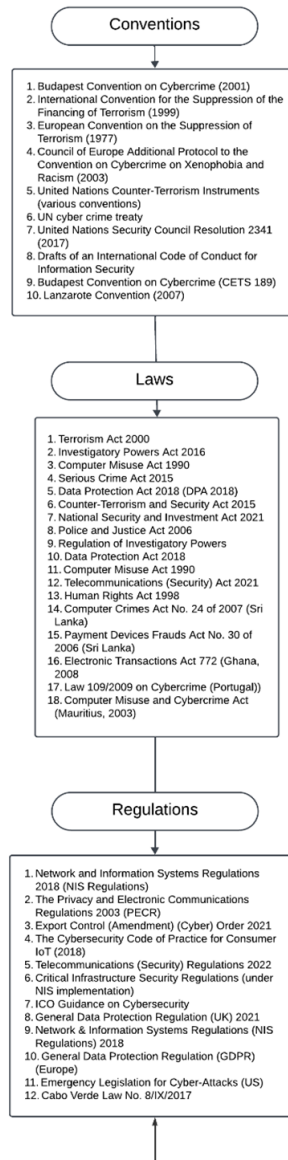
This mapping has been done to shed light on the manner in which the United Kingdom has structured its cybersecurity approach in the guise of a vast set of conventions, laws, legislation, and institutional arrangements. The UK has an international convention-driven multi-layered strategy and a very broad set of national legislation and regulations to protect critical infrastructure, combat cyber-enabled terrorism, and enhance overall cyber resilience.

Conversely, Sri Lanka relies mainly on a small range of international conventions—the most significant being the Budapest Convention—to serve as the basis for its local cybersecurity law. While this gives a starting point, some of Sri Lanka’s laws are outdated or limited in scope, and consequently, there are gaps in coverage compared to the UK’s stronger and more up-to-date legal base.

The UK strategy mirrors the need for the constant update and expansion of the cyber legislations to address emerging threats such as AI-led attacks, hybrid warfare, and safeguarding vital infrastructure. Further, the UK has developed different regulations and institutions that ensure harmonized national responses, which Sri Lanka lacks currently.

For Sri Lanka, the message is obvious: to fortify national security and be in a position to combat cyber terrorism, there needs to be closer alignment of domestic laws with global best practices, enlargement of the legal scope beyond the current acts, and stronger regulatory mechanisms and institutional frameworks.

Cyber Security Strategy



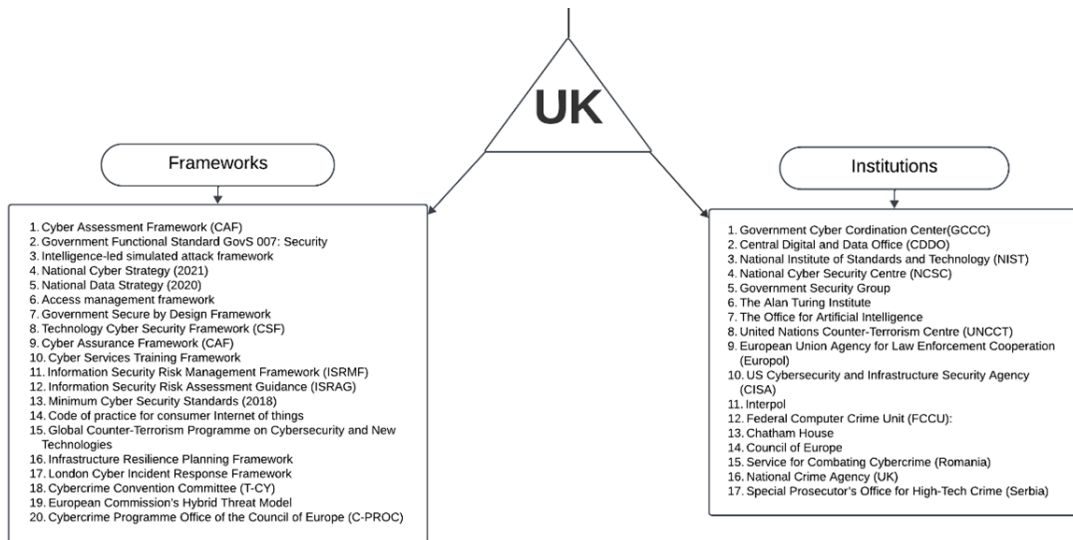


Figure 23: Comprehensive chart mapping the UK cybersecurity strategy

Conclusion

This research aimed to explore the legal and institutional preparedness of Sri Lanka to respond to cyber-terrorism, analyze the current legislation, assess the ability to enforce the law, and define the areas that are crucial. A series of gleaned insights via thorough analysis of documents, interviews with stakeholders. To begin with, the legal instruments used to establish the basis of Sri Lankan law on computer misuse, data privacy, and electronic commerce such as the Computer Crimes Act (2007), Online Safety Act (2024), Data Protection Act (2022), and Electronic Transactions Act (2006) offer general coverage of computer misuse, data privacy, and electronic commerce. But, all of them do not specify or aim cyber-terrorism, which makes prosecution and enforcement unclear. These laws have not dealt a lot with novel threat vectors including cryptocurrency-based financing, AI-based cognitive attacks, and autonomous weapon systems.

Second, there are constant challenges to the enforcement agencies. Criminal Investigation Department as well as other law enforcement agencies do not have enough technical competences, digital forensics capabilities, and institutionalized cross-border collaboration. According to stakeholders, skills shortage, old investigative practices and lack of cross-border cooperation were the major challenges when it comes to effective investigation and prosecution of cyber-terrorism incidents.

Multi-stakeholder views have highlighted the necessity to have better coordination between the government and the business, continued investment on cybersecurity training and infrastructure, and more explicit policies on balancing security and civil liberties. There is a strong focus on the need to constantly update the law in line with fast-changing cyber threats, alongside a strong focus on awareness campaigns and adoption of best practice across the industry. The implementation of a unified definition of cyber-terrorism under domestic law in order to remove interpretive loopholes.

Improving the digital evidence systems- defining the encrypted data access procedures, chain of custody, and court admissibility. The expansion of formal treaties on quick cross-border exchange of information, joint investigations and extraditions. Setting up of digital forensics, threat intelligence and incident response specific training programs and facilities. Incorporation of human rights to prevent any form of surveillance and enforcement to erode the basic freedoms.

Overall, though Sri Lanka has gone a long way in developing a rudimentary framework of cybercrime, it has to urgently streamline, update, and broaden the legal and institutional framework to tackle the specificity of cyber-terrorism. Through legislative modifications, enhancement of enforcement capacities and enabling ecosystems, Sri Lanka will be able to enhance its cyber resilience, improve the ability to protect critical infrastructure, and more effectively secure its people against the emerging threats of a digitally empowered adversary.